Applied Surface Science 612 (2023) 155776

Contents lists available at ScienceDirect

Applied Surface Science

journal homepage: www.elsevier.com/locate/apsusc

Full Length Article

Synergistic adsorption effect on Co_3O_4(1 1 0) surface to promote the ethanol sensing properties: Experiment and theory

Songlin Li [a], Chen Wang [a], Zhaohui Lei [a], Shanfu Sun [a], Jie Gao [b,c,*], Pengfei Cheng [a], Hai Wang [a,*]

[a] School of Aerospace Science and Technology, Xidian University, 266 Xifeng Road, Xi'an 710126, PR China
[b] Interdisciplinary Research Center of Biology & Catalysis, School of Life Sciences, Northwestern Polytechnical University, Xi'an 710072, PR China
[c] Research Institute of Industrial Technology, Zhengzhou University, Zhengzhou 450001, PR China

ARTICLE INFO

Keywords:
Co_3O_4
Crystal facets
Ethanol sensing
DFT
Adsorption configuration

ABSTRACT

A deep understanding of the relationship between the properties and structure of transition metal oxide semiconductor gas sensors is essential for optimizing sensor design. Comparing and analyzing the sensing properties of nanomaterials with different morphologies and crystal facets by combining the experiment analysis and the density functional theory (DFT) calculations could identify and understand the corresponding mechanisms at the atomic or ionic level. Here, grown Co_3O_4 nanosheet (Co_3O_4-NS) with exposed (1 1 1) facets, and Co_3O_4 nanorods (Co_3O_4-NR) with exposed (1 1 0) facets on alumina ceramic tubes were prepared and studied in sensing ethanol. The adsorption characteristics of ethanol on the sensor surface were analyzed by DFT calculation. The results demonstrated that the sensing performance of Co_3O_4-NR was superior to Co_3O_4-NS. The adsorption of hydroxyl groups in ethanol was better than the methyl groups on the surface. The oxygen and hydrogen atoms from the hydroxyl groups in ethanol had the lowest adsorption energies ($E_{ads} = -3.44$ eV) when they interact with the Co atom and the indirectly adjacent low-coordinate oxygen atoms on the (1 1 0) surface, respectively. The synergistic adsorption of target gases by exposed surface atoms induced superior sensing properties of the material.

1. Introduction

With the advancement of Internet of Things technology, new requirements are put forward for sensor technology [1,2], and gas sensors [3-5] play a pivotal role in the field [6,7]. Oxide semiconductor gas sensors present advantages of high sensitivity, low manufacturing cost, simple detection means, and convenient use [8-10]. Many factors can affect the gas-sensing properties of the materials, such as morphology [11], structure [12], surface defects [13], elements [14], BET surface area [15], or exposed crystal facets [16], etc. The difference in material morphology often exposes different crystal planes, which is the main factor causing the difference in material properties[17]. Materials of similar size and BET surface area have different reactivity due to differences in exposed crystal faces[18,19]. Equally, surface and interface properties play an important role in oxide semiconductor gas sensors [20-22]. Oxide semiconductor crystals exposed to different crystal facets lead to different sensing properties due to different surface atomic compositions, dangling bonds, and the existence of surface reconstruction[16,23]. In particular, oxide semiconductor sensors with different morphologies often have different adsorption properties owing to different facets[24-26], which reflects the relationship between structure and adsorption properties[27,28]. For example, the calculation by Rong et al. [29] shows that the (0 0 1) and (1 1 0) surfaces of WO_3 have completely different sensing properties. The NiO with foam structure obtained at 650 °C has excellent sensing performance due to the exposed (1 1 1) facets (presence of surface 3-coordinated unsaturated Ni atoms), opposite, the NiO with foamed and granular structures synthesized at 450 °C and 550 °C, respectively, have poor sensing characteristics owing to the exposed (0 2 2) facets [30]. Therefore, an increasing number of studies combine experimental [31] and theoretical calculations [32-34] to gain insight into the relationship between the sensing properties [35-37] of oxide semiconductor sensors exposed to different crystal planes [38-40].

Much effort has been devoted to obtaining high-sensitivity transition metal oxide semiconductor gas sensors [41-43]. Co_3O_4, as a p-type oxide semiconductor gas sensor nanomaterial, attracted much attention for

* Corresponding authors at: School of Aerospace Science and Technology, Xidian University, 266 Xifeng Road, Xi'an 710126, PR China (H. Wang).
E-mail addresses: jgao@nwpu.edu.cn (J. Gao), wanghai@mail.xidian.edu.cn (H. Wang).

https://doi.org/10.1016/j.apsusc.2022.155776
Received 26 September 2022; Received in revised form 1 November 2022; Accepted 16 November 2022
Available online 24 November 2022
0169-4332/© 2022 Elsevier B.V. All rights reserved.

the properties of anti-humidity properties and fast response and recovery time [44,45]. Previous studies have shown that the sensing properties of Co_3O_4 nanomaterials were closely related to their shape and exposed crystal facets [46,47]. For instance, Jian et al. [48] prepared Co_3O_4 with different crystal surfaces by hydrothermal method, and the result indicated that the surface of Co_3O_4(1 1 0) has a lower oxygen vacancy formation energy than the surface of (1 0 0), which leads to its high propane oxidation activity. Sun et al. [49] synthesized Co_3O_4 nanocube, and nanoplate exposed the surface of (1 0 0), and (1 1 2) by hydrothermal method, respectively. The NO_2 gas sensing properties of Co_3O_4(1 0 0) was higher than that of Co_3O_4(1 1 2), for a higher binding energy of NO_2 molecule onto the Co_3O_4(1 0 0). Oppositely, for smaller adsorption distances and greater charge transfer, the peroxidase-like activity of Co_3O_4(1 1 2) was higher than that of Co_3O_4(1 0 0). However, the relationship between the exposed crystal facets of Co_3O_4 and volatile organic compounds (VOC) gas sensing properties[50] still needs to be further investigated. The interaction of different functional groups in the VOC gas with the surface atoms and the adsorption configuration were rarely reported. Due to the complexity of the functional group, the joint action between different atoms of the functional group and multiple atoms on the sensor surface should not be neglected. By combining experimental characterization analysis and DFT calculation to study the VOC gas sensing behavior of different exposed surfaces of Co_3O_4, the corresponding mechanism can be understood at the macro and micro levels.

Herein, different morphologies exposed different crystal planes Co_3O_4 arrays (i.e., Co_3O_4 nanosheet (Co_3O_4-NS) with exposed (1 1 1) facets, and Co_3O_4 nanorods (Co_3O_4-NR) with exposed (1 1 0) facets) in-situ grown on alumina ceramic tubes were purposed. The Co_3O_4-NR and Co_3O_4-NS were widely characterized by various technologies: X-ray diffraction (XRD), X-ray photoelectron spectroscopy (XPS), field-emission scanning electron microscopy (FE-SEM), and high-resolution transmission electron microscopy (HR-TEM). The adsorption energies, the density of states (DOS), projected density of states (PDOS), and charge transfer were calculated to make insight into the relationship between the exposed surface and the sensing performance. The ethanol sensing performance of Co_3O_4-NR and Co_3O_4-NS were investigated by combining the experiments analysis and DFT calculations. We presented a detailed discussion on the effects of the adsorption configuration, exposed crystal facets, adsorption sites, and the synergistic effect of the sensor surface atoms on the sensing performance. This work not only provided a relationship between sensing properties and structure but also proposed a method to find and construct potential nanomaterials from the perspective of exposed crystal planes, surface atoms, and adsorption sites, which can also be used for other sensing materials.

2. Experimental

2.1. Preparation of Nano-structure Co_3O_4

Cobalt nitrate ($Co(NO_3)_2{\cdot}6H_2O$, 99%) was purchased from Sinopharm Chemical Reagent Co. Ltd. of China. Ammonium fluoride (NH_4F, 99%), urea ($CO(NH_2)_2$, 99%), and sodium citrate ($C_6H_5Na_3O_7{\cdot}2H_2O$, 99%) were purchased from Aladdin Ltd. All purchased chemicals are of analytical grade and can be used directly without further treatment. The Co_3O_4 nanomaterials in this paper were carried out based on the literature [51]. The preparation method was as follows: First, clean the alumina ceramic tube covering the gold electrode in Fig. 1 with alternating acetone and ethanol in the ultrasonic bath. Afterward, 10 mmol urea, 8 mmol ammonium fluoride, 2 mmol cobalt nitrate, and 0.03 mmol sodium citrate were dissolved in 36 ml deionized water (DI), stirring at room temperature for 30 min. In the next part, the prepared pink solution was added to a 40 ml Teflon liner, and a Teflon wire was passed through an alumina ceramic tube and suspended in the above pink solution. The Teflon inner chamber containing the above solution was put into a stainless-steel autoclave at 110 °C for 4 h and cooled to room temperature to obtain an alumina ceramic tube covered with a pink precursor. Then, the alumina ceramic tube covering the pink precursor was repeatedly washed with DI water, and ethanol and dried in the air environment of 80 °C for 6 h. Finally, the samples were annealed at 450∘C in the air atmosphere for 2 h to form the black Co_3O_4-NS. Under the same conditions, the growth time of the precursor body increased from 4 h to 9 h will obtain the Co_3O_4-NR. The preparation process of Co_3O_4 nanomaterials were shown in Fig. 1. The obtained prototype device can be directly used for gas sensing experiments.

2.2. Gas sensing property

The gas-sensing performance of the materials was measured by a static testing system (30 ± 5 RH%, 25 ± 1 °C) in two closed containers (1L) mixture with pure air and the target gas, respectively [15]. The lower concentration of the target gas was obtained by quantitatively pumping the purchased 1000 ppm target gas into the airtight chamber. Since the sensing material was grown directly in-situ on the alumina substrate, the process of producing and testing the sensor will be easier. As shown in Fig. 1, the gas sensor was obtained by soldering gold electrodes for the tube with sensing materials onto a pedestal with six nickel-plated probes. Additionally, the working temperature of the sensor was supplied by a Ni-Cr alloy coil. Different currents(I) provided by DC power (WON ODP-3033) result in different operating temperatures (T(K) = I(mA) × 2.5 + 224). The resistance of the gas sensor in different conditions was measured by a multimeter (Fluke 8846a) with a working voltage of 4.5 V. For P-type metal oxide semiconductor gas sensors, the resistance in reducing gases was higher than in the air condition. The response of the Co_3O_4 nanoarray gas sensor in this paper was defined as S = Rg/Ra, where Rg and Ra are the resistance values of the sensor in the target gas and air, respectively. Response time and recovery time were the time it takes for the resistance to rise or fall to 90% of the total resistance change, respectively [52].

2.3. Calculation details

Theoretical calculations were performed using the first-principles software VASP (Vienna Ab initio Simulation Package) [53,54]. To correct the large self-interaction error of the 3d electron-electron correlation effect of Co, density functional theory (DFT)+U was used and the

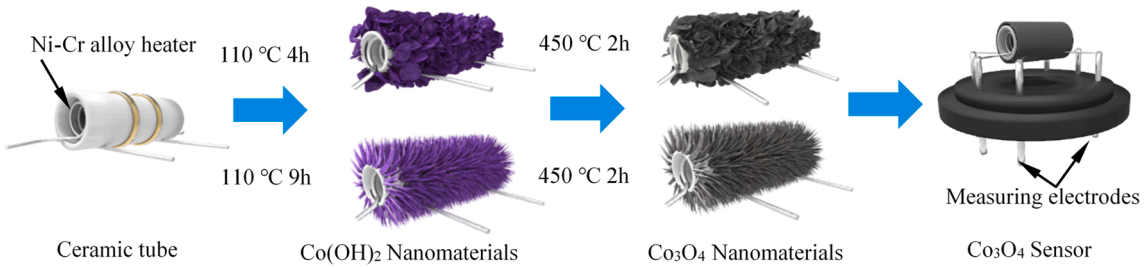

Fig. 1. Schematic diagram of the in-situ growth process of Co_3O_4 nanomaterials.

U_{eff} = U - J value was set to 3.0 eV [55]. The exchange-correlation functional was GGA-PBE (Generalized gradient approximation-Perdew Burke Ernzerhof) [56,57] to describe the interactions between gas molecules and the Co_3O_4 surface. The Co_3O_4 was antiferromagnetic at low temperatures and changed to a paramagnetic state when the temperature was higher than 40 K. So, we used the ferromagnetic spin state to model the Co_3O_4 [58,59]. Spin polarization was applied in all calculations unless otherwise specified. The projector augmented wave (PAW) potentials were used to describe the electron-ion interactions [53,54]. The cutoff energies were 400 eV for all of the computations used for the expansion of plane waves. The electronic energy difference was converged to 1E-5 eV in each self-consistent field (SCF). The ionic relaxation stopped when the norms of all the forces were smaller than 0.02 eV/Å. Brillouin zone integration of Co_3O_4 bulk, the surface of Co_3O_4(1 1 1), and Co_3O_4(1 1 0) were calculated by 6 × 6 × 6, 3 × 3 × 1, and 3 × 2 × 1 Monkhorst-Pack k-point meshes, respectively [60]. To prevent interactions between the slabs, a 15 Å vacuum was inserted in the z-direction. The top two and three layers were fully relaxed for Co_3O_4(1 1 0) and Co_3O_4(1 1 1), respectively. The other layers were fixed to model the bulk phase during structural optimizations. The adsorption energy [61] was defined by equation (1),

$$E_{ads} = E_{Surface/Gas} - E_{Surface} - E_{Gas} \quad (1)$$

where $E_{Surface/Gas}$ and $E_{Surface}$ are total energies of the slab module with and without gas, and E_{Gas} is the total energy of adsorbed gas molecules. The charge density difference [62] (CDD) is calculated by equation (2),

$$\Delta\rho = \rho_{Surface/gas} - \rho_{Surface} - \rho_{Gas} \quad (2)$$

where $\rho_{Surface/gas}$ and $\rho_{Surface}$ are the charge density of the slab module with and without gas, respectively, and ρ_{Gas} is the charge density of the adsorbed gas.

3. Results and discussion

3.1. Characterization of Co_3O_4 nanomaterials

The micromorphology of sensors Co_3O_4-NR and Co_3O_4-NS were carried out by FE-SEM and HR-TEM. A well-grown Co_3O_4-NR array on the sensor surface could be seen from the top and side views in Fig. 2a, and c. Further structural characteristics were obtained by HR-TEM, as shown in Fig. 2e. The lattice arrangement and the corresponding FFT pattern in Figure S1 and Fig. 2e inset suggested that the lattice fringe spaces on Co_3O_4-NR was 0.202 nm which was corresponding to the (4 0 0) surface of Co_3O_4. The above information suggested that the sensor of Co_3O_4-NR primarily exposed the surface of {1 1 0} (perpendicular to the surface of (4 0 0), and (2 2 2)). Moreover, the side and top views of the in-situ growth of Co_3O_4-NS on the sensor surface could be seen in Fig. 2b, and d, respectively. The HR-TEM image of the Co_3O_4-NS arrays was shown in Fig. 2f, and the corresponding lattice arrangement and FFT pattern were shown in the inset of Fig. 2f and Figure S2. The HR-TEM picture and FFT pattern revealed that the sensor of Co_3O_4-NS primarily exposed the surface of {1 1 1}. The crystal surface exposures were consistent with those in the literature [14].

XRD (Rigaku D/Max-2550 diffractometer) patterns of Co_3O_4-NS and Co_3O_4-NR with Cu-Kα radiation (0.15418 nm) within 20°–80° was shown in Fig. 3a, and the (2 2 0), (3 1 1), (4 0 0), (5 1 1), and (4 4 0) planes were attributed to the standard position of cubic spinel Co_3O_4 (PDF#78-1969). There were no other impurity peaks, demonstrating the high purity of the prepared nanomaterials. Moreover, to obtain the chemical status and surface composition of synthesized material, XPS patterns of Co_3O_4-NR and Co_3O_4-NS were shown in Fig. 2b. Throughout the full XPS survey spectra, only C, Co, and O elements were observed, confirming that all materials were pure cobalt oxides. The Co 2p XPS spectra of Co_3O_4-NR and Co_3O_4-NS were shown in Fig. 3c. The two main peaks in the Co 2p XPS spectra were attributed to Co $2p_{3/2}$ (780 eV) and Co $2p_{1/2}$ (795 eV), respectively. The peak fitting of Co $2p_{3/2}$ could obtain two peaks of Co^{3+} and Co^{2+} with the binding energy of 779.73 eV and 781.20 eV, respectively. The O 1s XPS spectra of Co_3O_4-NR and Co_3O_4-NS were shown in Fig. 3d. The peak fitting of O 1s could obtain two peaks of the lattice oxygen (O_L) with the binding energy of 529.87 eV and low coordinated oxygen species (O_C) with the binding energy of 531.06 eV. The high concentration of adsorbed and low coordinated oxygen species on the sensor surface could directly promote the sensing performance. As shown in Table 1, the Co_3O_4-NR had the highest O_C/O_L ratio (0.66), followed by Co_3O_4-NS (0.53). Moreover, the $Co^{3+}/Co+$ ratio of Co_3O_4-NR and Co_3O_4-NS was 2.21 and 2.81, respectively. A lower Co^{3+}/Co^{2+} ratio indicated more O_C in Co_3O_4-NR [14,63].

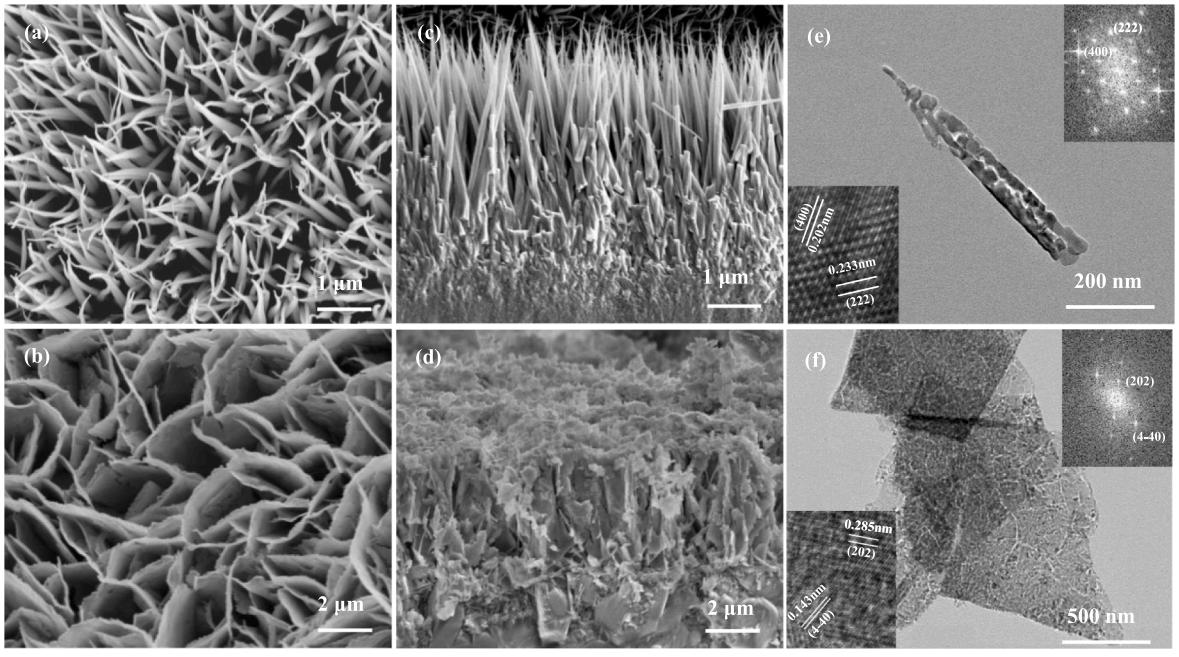

Fig. 2. (a-d) FE-SEM images of Co_3O_4-NR and Co_3O_4-NS, (e-f) HR-TEM images of Co_3O_4-NR and Co_3O_4-NS. Insets were the corresponding high-resolution images and FFT patterns.

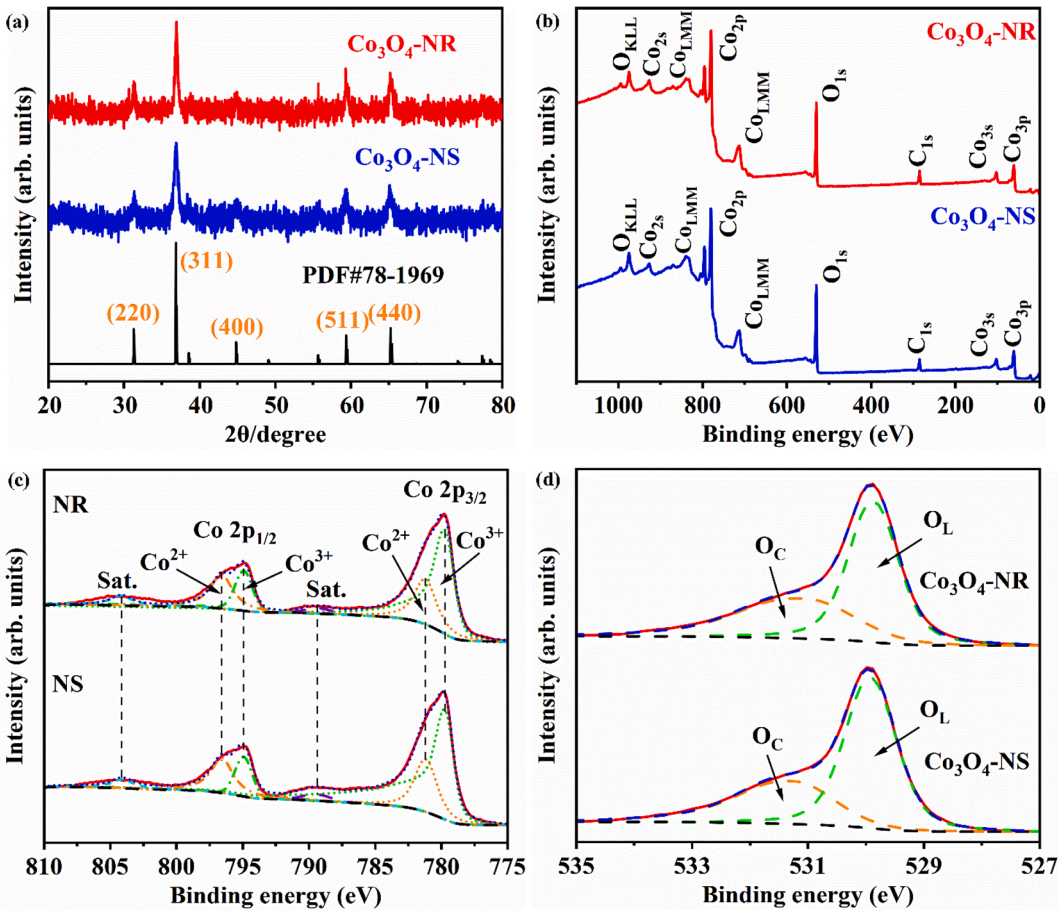

Fig. 3. (a) XRD pattern of Co_3O_4-NS and Co_3O_4-NR, (b) XPS survey spectra and corresponding, (c) Co 2p, (d) O 1s XPS.

Table 1
XPS results of prepared Co_3O_4 nanomaterials.

Material	Binding energy (eV)				Co^{3+}/Co^{2+}	O_C/O_L
	$Co^{3+}2p_{3/2}$	$Co^{2+}2p_{3/2}$	O_L	O_C		
Co_3O_4-NR	779.75 (447479)	781.18 (157911)	529.92 (181759)	531.26 (95504)	2.21	0.66
Co_3O_4-NS	779.73 (330907)	781.20 (149563)	529.87 (163637)	531.06 (108476)	2.81	0.53

3.2. Gas-sensing performance of Co_3O_4 nanosheets and nanorods

The gas-sensing performance was conducted by the resistances of Co_3O_4-NS and Co_3O_4-NR sensors in the air and target gas. As p-type metal oxide semiconductors (MOSs), the response of the Co_3O_4 gas sensor was defined as its resistance value in the target gas (Rg) divided by the resistance value in the air (Ra). The sensing sensitivity of the MOSs gas sensor was closely related to their operating temperature, so the response values of the sensors with different morphologies in 100 ppm ethanol gas were measured at different temperatures. As seen in Fig. 4a, the response of Co_3O_4-NS and Co_3O_4-NR sensors at different temperatures was presented as a typical "volcano" shape. Compared with the Co_3O_4-NS sensor in Fig. 4a, the Co_3O_4-NR sensor exhibit the highest response values (21.46) to 100 ppm ethanol at their optimum operating temperature. Furthermore, Fig. 4b presented the time-resistance characteristics of the Co_3O_4-NR sensor to 100 ppm ethanol at 150 °C (When the resistance value is stable, it was placed in and away from the ethanol gas environment).

The dynamic response-recovery curve expressed that the resistance of the Co_3O_4-NR sensor rises rapidly in the ethanol environment and decreases rapidly in the air ambient, showing obvious p-type characteristics. And the corresponding response and recovery times were 137 s and 128 s, respectively. Furthermore, as shown in Fig. 4c, the response value of Co_3O_4-NR and Co_3O_4-NS sensor to 100 ppm of other six target gases (acetone, formaldehyde, methanol, acetic acid, isopropanol, ethylbenzene) were obtained at their respective optimum operating temperature. The radar diagram presented that the Co_3O_4-NR sensor has the highest response value to ethanol, and the Co_3O_4-NR sensor has the best gas sensing sensitivity to all the eight gases than the Co_3O_4-NS sensor. Additionally, Fig. 4d shown the response ratios ($S_{ethanol}/S_{others}$) of the Co_3O_4-NR and Co_3O_4-NS sensor to ethanol versus other gases, in which the higher the ratio, the better selectivity. As seen in Fig. 4d, the response ratios to all seven gases of the Co_3O_4-NR sensor were higher than the Co_3O_4-NS sensor, suggesting a higher potential selectivity.

To further observe the characteristics of the Co_3O_4-NR sensor. The time-response properties of the Co_3O_4-NR sensor at different ethanol concentrations were shown in Fig. 5a. The response value of the Co_3O_4-NR sensor gradually increased with the increasing concentration of ethanol. The dynamic response-recovery curve at the ethanol concentration from 2.5 ppm to 200 ppm demonstrated an excellent response and recovery characteristics of the Co_3O_4-NR sensor. Fig. 5b presented the dot-line response graph of the Co_3O_4-NR sensor to different concentrations of ethanol, when the concentration of ethanol was less than 100 ppm, the response value increases linearly as the ethanol concentration increased. While the concentration was above 100 ppm, the response value slowly increases as the ethanol concentration increased, which means that the sensor gradually saturated. The linear fitting curve within 2.5–100 ppm was expressed as $S = 0.204X + 0.797$ ($R2 = 0.995$), in which S, X, and R2 are the response values, ethanol concentration,

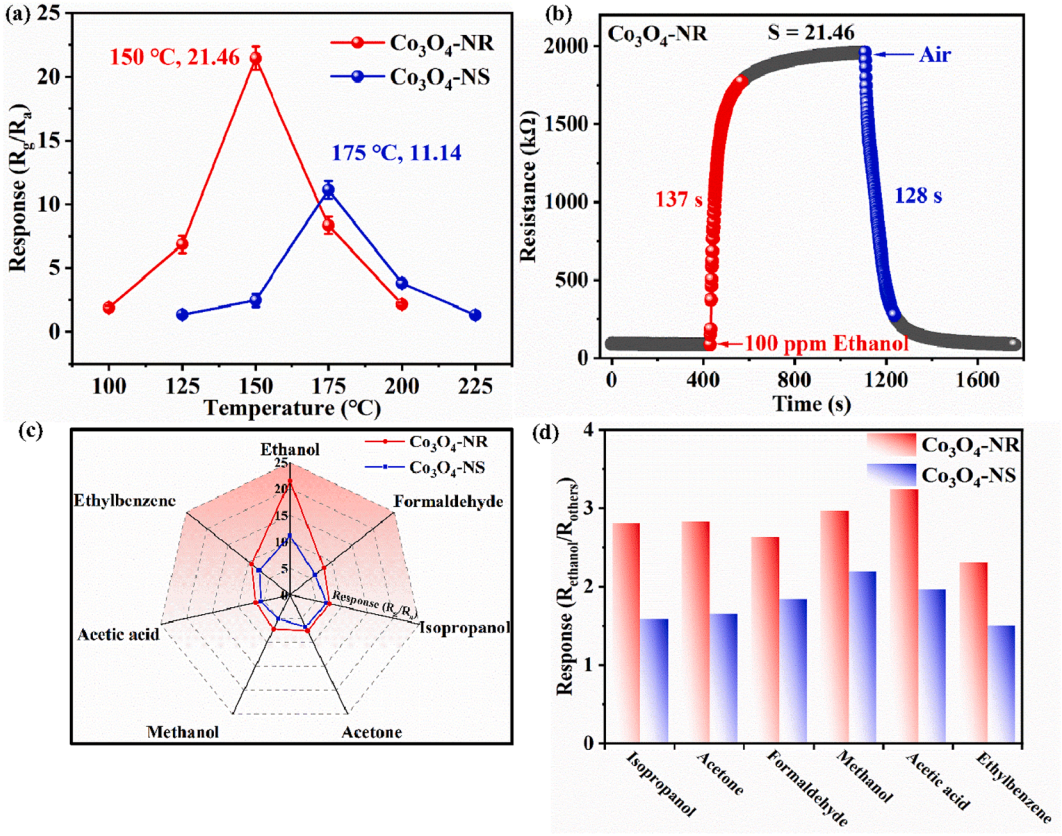

Fig. 4. (a) Responses of Co_3O_4-NR and Co_3O_4-NS to 100 ppm ethanol at different working temperatures, (b) Response-recovery time curve of the Co_3O_4-NR sensor to 100 ppm ethanol at 150 °C, (c) Responses of obtained sensors towards 100 ppm different gases at their optimal temperature, (d) The response ratios of obtained sensors to ethanol versus other gases.

and the linearity of the fitted curve, respectively. The linearity closed to 1.0, suggesting that the Co_3O_4-NR sensor had high linearity in response to 2.5–100 ppm ethanol. The short/long-term stability of the Co_3O_4-NR sensor was plotted in Fig. 5c and d. As seen in Fig. 5c, the similar dynamic response-recovery cycles proposed that the Co_3O_4-NR sensor illustrated stable repeatability. As plotted in Fig. 5d, the response values slightly fluctuated around 21, demonstrating that the Co_3O_4-NR sensor owned great long-term stability.

3.3. Bulk and surface properties of Co_3O_4

The structure information of Co_3O_4 was obtained by PDF#78-1969 as shown in Fig. 6a and Table 2. The red represents the oxygen atom, the blue represents the Co^{3+}, and the pink represents Co^{2+}. The bulk structure in MAGNDATA [64,65] (A Collection of magnetic structures with portable cif-type files) was a = b = c = 8.065 Å, α = β = γ = 90.0°. The error of lattice parameters between calculated and database was within 0.5% in Table 2. Combined with XRD data and structural parameters indicated the high purity of the materials and the rationality of the parameters used in calculation.

The module for Co_3O_4-NR and Co_3O_4-NS exposed the surface of (1 1 0) and (1 1 1) respectively, were established with supercells of 2 × 2 × 2, and $\sqrt{3}\times\sqrt{3}$ × 2 unit cells, respectively, as shown in Fig. 6b-e. According to the previous work [55], Co_3O_4 (1 1 0) had two different surface structures labeled A and B, respectively. Surface-A exposed two types of metal atoms (Co^{2+} and Co^{3+}) and one kind of oxygen atom (3-fold coordinated O_{3c}). Surface-B exposed two kinds of oxygen atoms (2-fold coordinated O_{2c} and O_{3c}) and one type of metal atom (4-fold coordinated Co^{3+}). We used the B-type of Co_3O_4(1 1 0) surface to simulate the sensor surface because B-type was more stable than A-type. Equally, the most stable surface of Co_3O_4(1 1 1) exposed one type of atom (O_{3c}) and two kinds of metal atoms (3-fold coordinated Co^{2+} and Co^{3+}). Each of the considered slabs consists of 84 atoms ($Co_{36}O_{48}$). After structure optimization, the bond length of Co^{2+} and O_{3c} on the surface of Co_3O_4(1 1 1) changed from 1.929 Å to 1.790 Å and 1.771 Å. The bond length of Co^{3+} and O_{2c} on the surface of Co_3O_4(1 1 0) changed from 1.929 Å to 1.707 Å. The bond length of Co^{3+} and O_{3c} changed from 1.916 Å to 1.848 Å. Moreover, the Co 2p and O 1 s XPS spectra in Fig. 3c and d also indicated that the materials of Co_3O_4-NR (exposed surface of (1 1 0)) had more low-coordination oxygen than the materials of Co_3O_4-NS (exposed surface of (1 1 1)), which was consistent with our surface model in the TEM. Furthermore, the bond length of the low-coordinated oxygen and cobalt varied significantly on the surface of Co_3O_4(1 1 0), indicating that the adsorption sites of the low-coordination oxygen might have higher activity.

3.4. Adsorption properties of Co_3O_4 (1 1 1) and Co_3O_4 (1 1 0)

To further discuss the atomic configurations and adsorption characteristics for the adsorption of ethanol on the surface of Co_3O_4(1 1 0) and Co_3O_4(1 1 1) were shown in Fig. 7. The Bader charge transfer [66], and adsorption distance for ethanol molecule on the surface of Co_3O_4(1 1 0) and Co_3O_4(1 1 1) were calculated in Table 3. As shown in Fig. 6c, Co_3O_4(1 1 0) exposed two types of oxygen atoms (O_{2c} and O_{3c}) and one kind of metal atom (4-fold coordinated Co^{3+}). As seen in Fig. 7a-j, we calculated five adsorption configurations of the ethanol molecule on the Co_3O_4(1 1 0). Among them, S1 and S2 symbolize the hydroxyl group of the ethanol molecule that was adsorbed between Co^{3+} atoms and directly/indirectly adjacent O_{2c}. S3 and S4 denote the hydroxyl group of the ethanol molecule that was adsorbed between Co^{3+} atoms and directly/indirectly adjacent O_{3c}. S5 represents the methyl group of the ethanol molecule that was adsorbed on top of O_{2c}. The adsorption

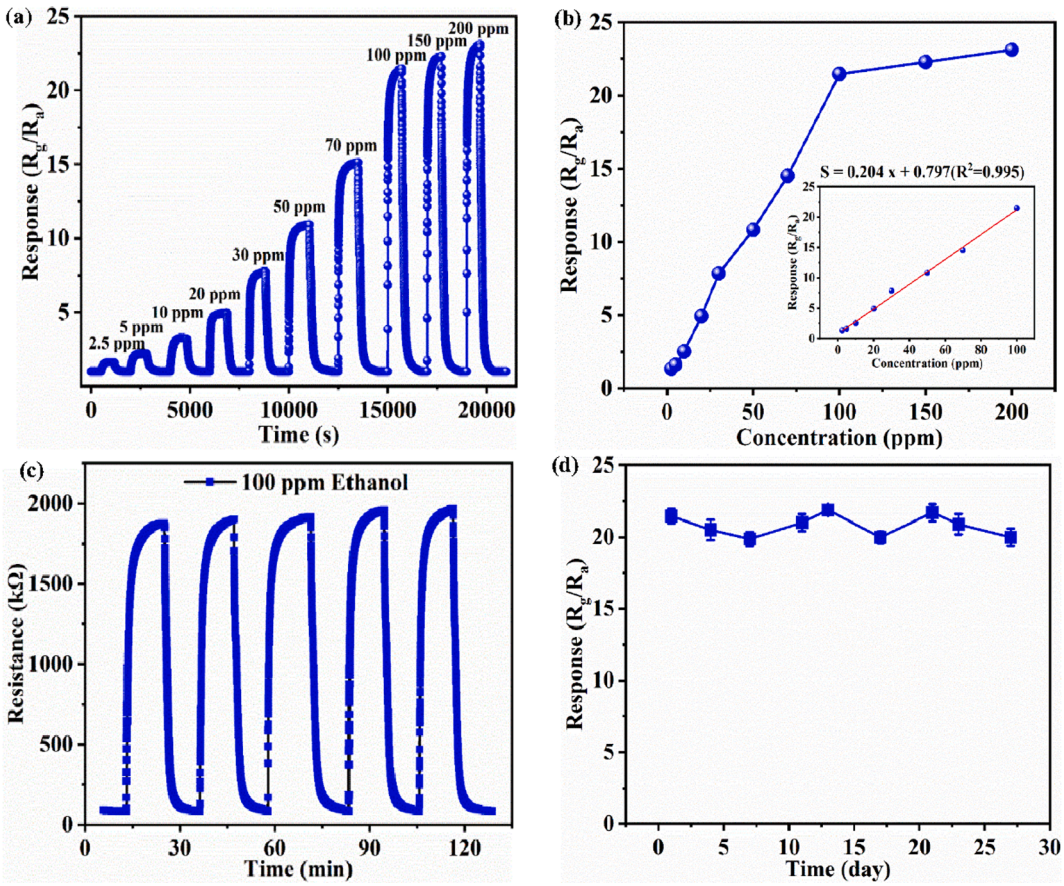

Fig. 5. (a) Dynamic response-recovery patterns of the Co_3O_4-NR sensor to ethanol in the concentration range of 2.5–200 ppm, (b) Dot-line response graph and linear curves of the Co_3O_4-NR sensor to different concentrations of ethanol gas (c) Dynamic response-recovery patterns of Co_3O_4-NR sensor to 100 ppm ethanol at 150 °C, (d) Long-term stability of the Co_3O_4-NR sensor to 100 ppm ethanol at 150 °C.

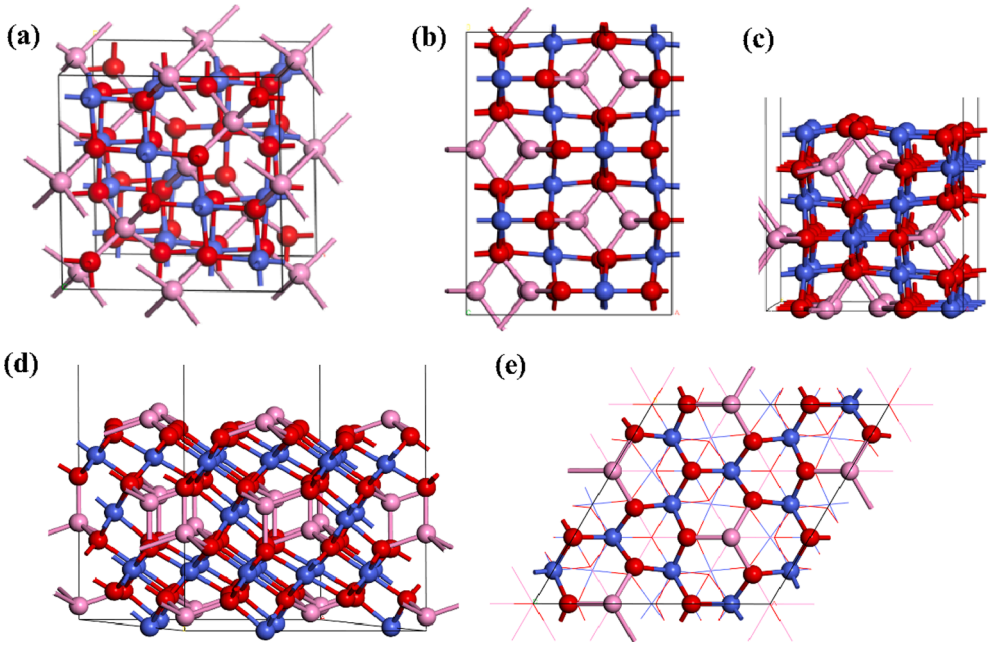

Fig. 6. (a) structure information of Co_3O_4 bulk (the red represents the oxygen atom, the blue represents the Co^{3+}, and the pink represents Co^{2+}), (b, c) the top and side view of Co_3O_4(1 1 0)-B after optimized, (d, e) the top and side view of Co_3O_4(1 1 1) after optimized.

Table 2
The crystal structure information for Co_3O_4, in the MAGNDATA, PDF card and DFT calculations.

	Database	PDF card	DFT+U
a (Å)	8.065	8.085	8.061
b (Å)	8.065	8.085	8.061
c (Å)	8.065	8.085	8.061
α (deg)	90.0	90.0	90.0
β (deg)	90.0	90.0	90.0
γ (deg)	90.0	90.0	90.0

configuration of ethanol on the surface of Co_3O_4(1 1 1) was shown in Fig. 7k-l.

Based on the adsorption energies calculated in Table 3, the –OH in ethanol adsorbed on the surface of Co_3O_4(1 1 0) shows smaller energies of −3.44, −2.91, −2.18, and −2.85 eV for sites S1-S4, respectively, compared with that for site S5 of −0.42 eV. Comparing the adsorption sites of S1-S4 on Co_3O_4(1 1 0), the calculated adsorption energies show that the hydroxyl group of the ethanol molecule was more easily adsorbed on the surface than the methyl group. The synergistic adsorption effect between the low-coordinated O_{2c} and Co on the Co_3O_4(1 1 0) surface was more favorable for the hydroxyl adsorption of the ethanol molecule. The ethanol adsorption energy on the Co_3O_4(1 1 1) surface was −0.53 eV, which was larger than that for Co_3O_4(1 1 0) surface. Comparing the adsorption energies on Co_3O_4(1 1 0) and Co_3O_4(1 1 1), the ethanol molecule was more easily adsorbed on the surface of Co_3O_4(1 1 0). Based on the adsorption distance calculated in Table 3, –OH in ethanol tends to form H-O_{2c} bonds and Co^{3+}-O bonds with the surface of Co_3O_4(1 1 0) minimum lengths of 1.56 Å and 1.95 Å, respectively, which were smaller than Co^{2+}-O bonds found for the Co_3O_4(1 1 1) surface of 2.26 Å. The synergistic adsorption effect caused a smaller adsorption distance between the ethanol molecules and the surface of Co_3O_4(1 1 0).

Comparing the adsorption site of S1-S5 on Co_3O_4(1 1 0), the calculated adsorption distances show that the hydroxyl of the ethanol molecule was more easily adsorbed than the methyl. The Oxygen and hydrogen atoms in the hydroxyl of ethanol form stable adsorption on the

Table 3
Ethanol adsorption configuration on Co_3O_4(1 1 0) and Co_3O_4(1 1 1), and calculated corresponding parameters of the adsorption energy (Eads), adsorption distance, and charge transfer (Q).

Site	Co_3O_4(1 1 0)					Co_3O_4(1 1 1)
	S1	S2	S3	S4	S5	
Adsorption energy (eV)	−3.44	−2.91	−2.18	−2.85	−0.42	−0.53
Distance (Å)	1.59	2.15	1.56	1.95	2.52	2.26
Transferred electron (e)	−0.11	−0.10	−0.12	−0.10	0.006	−0.03

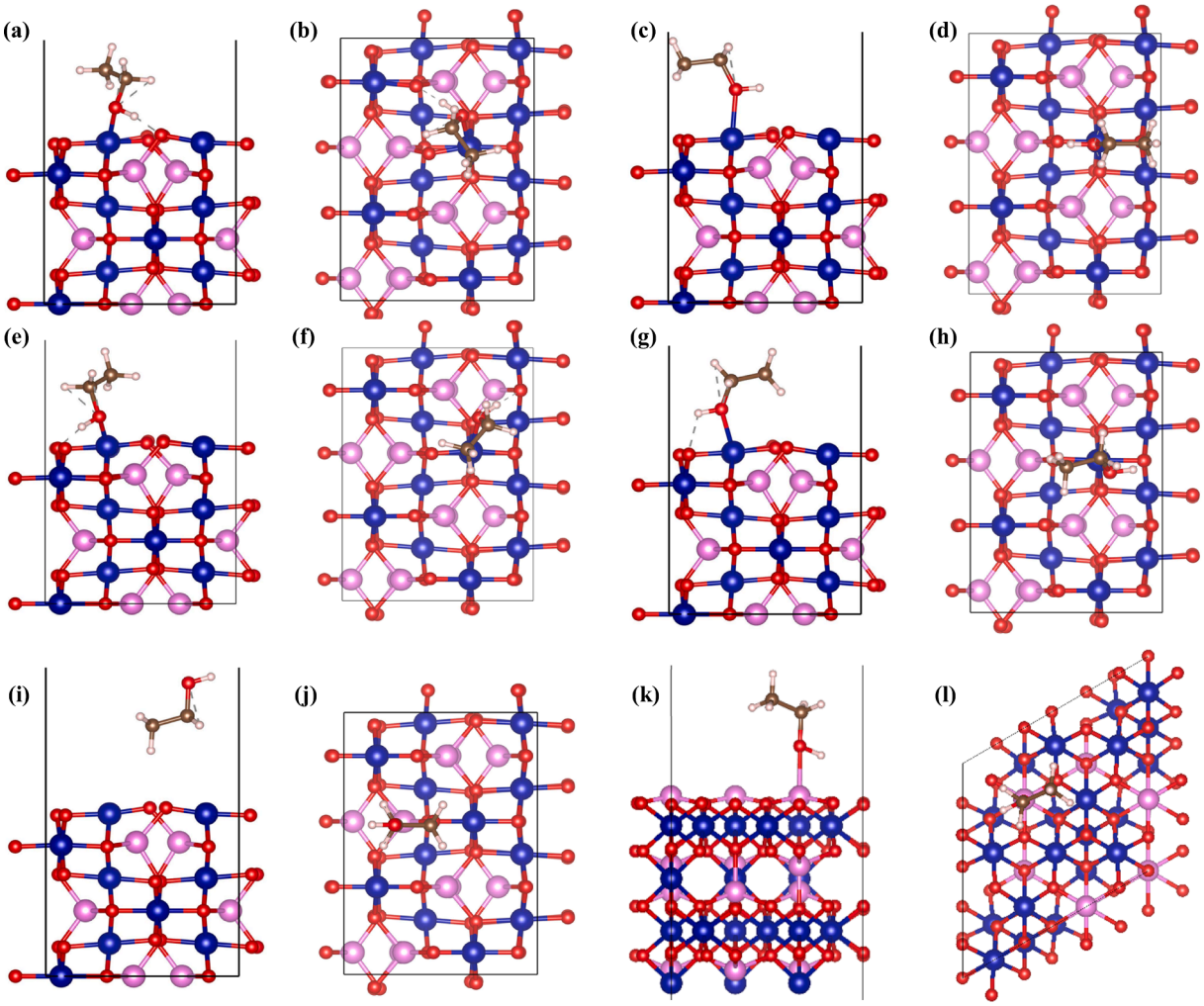

Fig. 7. Structure information of ethanol adsorb on the surface of Co_3O_4 (the red represents the oxygen atom, the blue represents the Co^{3+}, and the pink represents Co^{2+}, the white represents the hydrogen atom), (a - j) the top and side view of ethanol adsorbed on the surface of Co_3O_4(1 1 0) on site S1-S5 after optimized, (k, l) the top and side view of ethanol adsorbed on the surface of Co_3O_4(1 1 1) after optimized.

Co_3O_4(1 1 0) surface of Co^{3+} and O (O_{2c} and O_{3c}), respectively. The adsorption of the hydrogen atom in the methyl of ethanol on the Co_3O_4(1 1 0) surface of O_{2c} was weaker than the synergistic adsorption characteristics formed by the hydroxyl group. Besides, for the synergistic adsorption effect, the ethanol adsorption illustrated a greater charge transfer of −0.12 |e| to −0.10 |e| for Co_3O_4(1 1 0) surface of site S1-S4, respectively, compared with that of Co_3O_4(1 1 1) for −0.03 |e|. The above results disclosed a chemical adsorption behavior of hydroxyl in ethanol with Co and O_{2c} on the Co_3O_4(1 1 0) surface and the highly sensitive sensing properties of ethanol gas. As shown in Table 3 and Fig. 7, the calculated adsorption energy, Bader charge transfer, and adsorption distance suggest that the presence of a synergistic adsorption effect results in excellent sensing properties of Co_3O_4 (1 1 0), which aligned with the results in Fig. 4.

3.5. Project density of states analysis

To analyze the electronic property of ethanol adsorbed on the surface of Co_3O_4(1 1 0) and Co_3O_4(1 1 1), the DOS and PDOS for the Co_3O_4(1 1 0) and Co_3O_4(1 1 1) systems were shown in Fig. 8 and Fig. 9. The red curve represents the DOS of Co_3O_4(1 1 0) and Co_3O_4(1 1 1), and the blue line represents the DOS of the surface after the adsorption of the ethanol

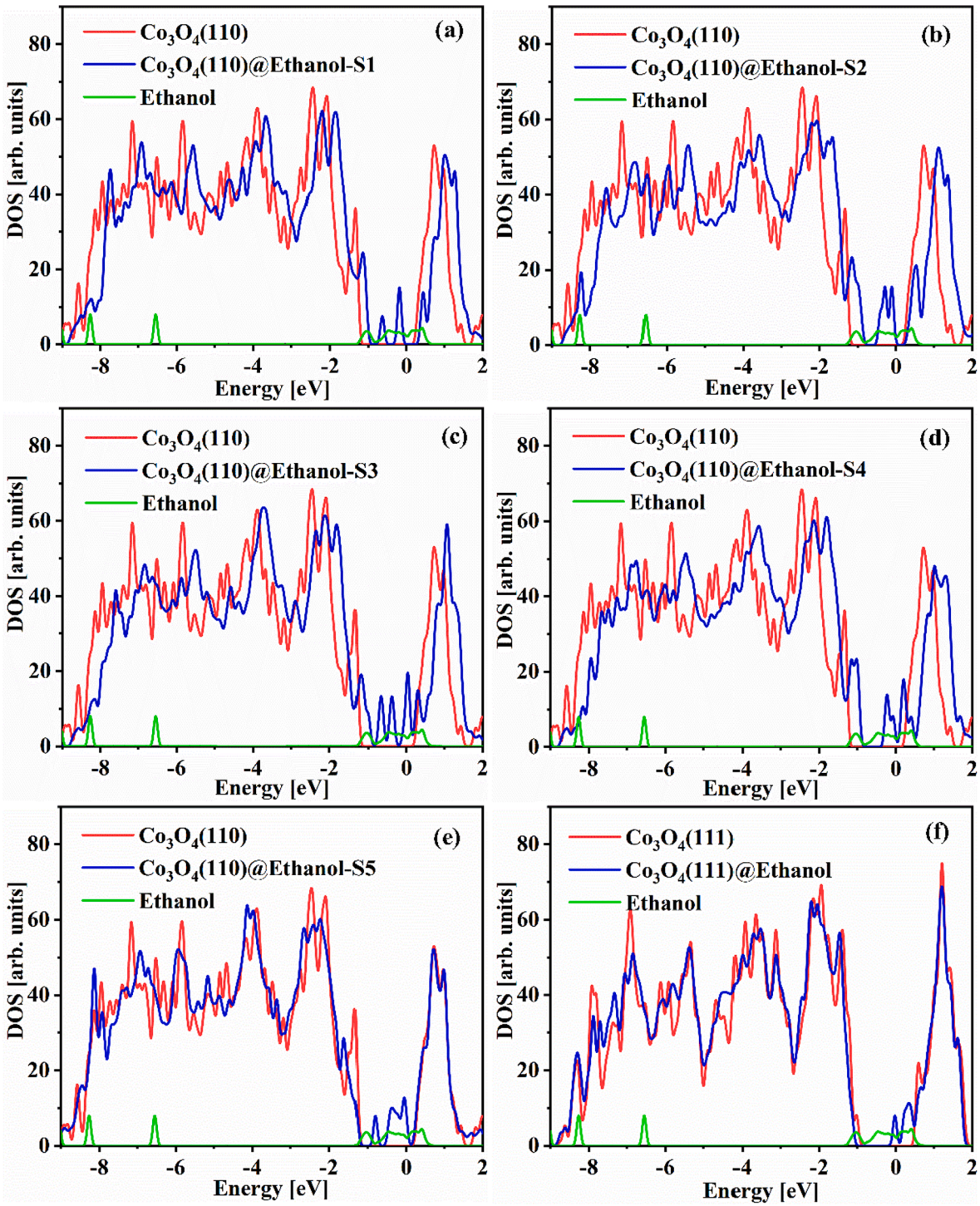

Fig. 8. DOS of Co_3O_4(1 1 0) and Co_3O_4(1 1 1) surface before and after ethanol adsorption.

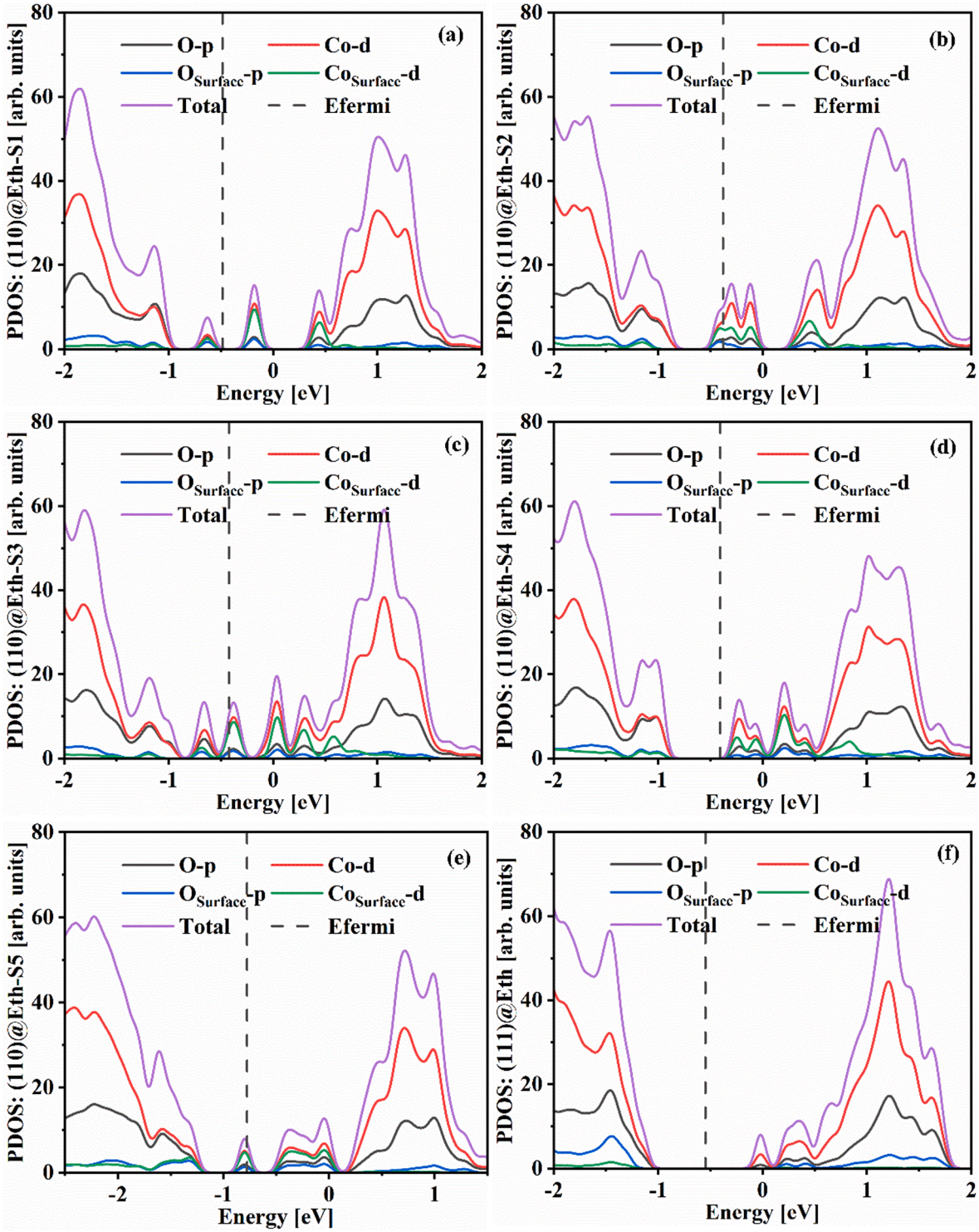

Fig. 9. PDOS of Co_3O_4(1 1 0) and Co_3O_4(1 1 1) surface after ethanol adsorption.

molecule. As seen in Fig. 8a-e, the bandgap for Co_3O_4(1 1 0) systems had significant changes before and after the adsorption of the ethanol molecule. Moreover, the DOS of Co_3O_4(1 1 0) systems moves to higher energy when ethanol adsorption on the site S1-S4 compared with that for the site of S5. Combined with the previous calculation results of adsorption energy and charge transfer, it was suggested that the presence of the synergistic adsorption effect results in the adsorption of the hydroxyl group in ethanol on the Co_3O_4(1 1 0) surface was more likely to cause electronic state changes than methyl groups. The introduction of the ethanol molecule leads to the emergence of new electronic states in the forbidden band of Co_3O_4(1 1 0). But as shown in Fig. 8f, the DOS of Co_3O_4(1 1 1) systems has hardly changed before and after the adsorption of the ethanol molecule. The bandgap of the Co_3O_4(1 1 1) system changed a little before and after ethanol adsorption, and the surface states were not found in the forbidden band, demonstrated a lower adsorption property than Co_3O_4(1 1 0).

To further analyze the electronic characteristics of the sensor, the electronic characteristics of different atoms with different orbitals were calculated as shown in Fig. 9. The grey and red curves represent the p-orbital of O and the d-orbital of Co atom in Co_3O_4, respectively. The blue

and green curves denote the p-orbital of O and the d-orbital of Co on the surface of Co_3O_4, respectively. The purple curve indicates the total density of states of Co_3O_4 after ethanol adsorption. Comparing the PDOS of ethanol adsorption on the surface of Co_3O_4(1 1 0), the surface states in the forbidden band were mainly contributed by the d-orbital of Co and the p-orbital of O. The almost overlapping red and green curves, grey and blue curves in Fig. 9a-e indicated that the surface states in the forbidden band were mainly caused by the Co and O atoms on the surface of Co_3O_4(1 1 0) which were especially pronounced when ethanol was adsorbed on the S1 site of Co_3O_4(1 1 0). The synergistic adsorption effect between the surface atoms and the hydroxyl groups greatly changed their electronic states. However, the PDOS of ethanol adsorption on the surface of Co_3O_4(1 1 1) has a small change, and the DOS contribution of Co and O on the surface of Co_3O_4(1 1 1) was submerged in the contribution of the bulk phase of Co_3O_4. The appearance of surface states may reduce the mobility and thus surface conductivity. The above analysis corresponds to the increase in the resistance of Co_3O_4 caused by the introduction of ethanol gas shown in Fig. 4. Combined with the above analysis, it was shown that the synergistic adsorption effect of the –OH in ethanol and the surface atoms effectively alters the characteristics of Co_3O_4(1 1 0) which was similar to the previous test results of ethanol gases adsorbed on the surface of Co_3O_4-NR and Co_3O_4-NS.

3.6. Charge density difference analysis

The CDD for the ethanol adsorption on the surface of Co_3O_4(1 1 0) and Co_3O_4(1 1 1) was shown in Fig. 10. The blue and yellow areas represent depletion and accumulation of charges, respectively. The gain and loss of electrons between the ethanol molecule and the surfaces of Co_3O_4(1 1 0) and Co_3O_4(1 1 1) could be observed in it. The charge transfer between the ethanol molecule and the surface of Co_3O_4(1 1 0) was significantly higher than that with the surface of Co_3O_4(1 1 1), indicating that the ethanol molecule was more likely to interact with the Co_3O_4(1 1 0) surface. As shown in Fig. 10a-d, comparing the five adsorption sites on the surface of Co_3O_4(1 1 0), the existence of the synergistic adsorption effect results in the charge has a more significant change in the electronic state between the hydroxyl groups of ethanol molecules interacting with the surface of Co_3O_4(1 1 0). The above analysis expressed that the presence of the synergistic adsorption effect results in the hydroxyl group of the ethanol molecule was more easily adsorbed than the methyl group, which was consistent with the above calculations. Moreover, compared the adsorption sites of S1 and S3 with S2 and S4 in Fig. 10a-d, the adsorption of hydroxyl on the Co atom and the O atom indirectly adjacent to this Co atom was stronger than that on the Co and O atom directly adjacent to this Co atom. Compared with the surface of Co_3O_4(1 1 1), the surface of Co_3O_4(1 1 0) has a large number of such S1 and S3 adsorption sites. The above analysis shows a strong sensitivity for the synergistic adsorption effect between the surface of Co_3O_4(1 1 0) and ethanol.

3.7. Sensing mechanism

The change in conductivity of the sensor was primarily responsible for the different response values of Co_3O_4 nanorods and nanosheets. As previously stated, the variation in conductivity of oxide semiconductor gas sensors was caused by the interactions between the target gas and the sensor surface. Moreover, the gas adsorption properties played a crucial role in gas sensing. Fig. 11 displayed a schematic diagram of the sensing mechanism of the Co_3O_4 nanorods and nanosheet sensors with ethanol gas. As a P-type transition metal oxide semiconductor gas sensor, the holes in the Co_3O_4 material will act as the main carrier in the semiconductor. The sensor exhibits a lower resistance value in air conditions. When the sensor was exposed to ethanol gas, the resistance of the sensor increased due to the interaction of the ethanol molecules with the sensor surface. The sensor's resistivity changes when electrons were exchanged between ethanol molecules and the sensor surface. Carrier concentration had a strong influence on semiconductor resistivity. The relationship between the resistivity of a semiconductor and the carrier concentration was shown in equation (3).

$$\rho = \frac{1}{\sigma} = \frac{1}{e(\mu_n n + \mu_p p)} \tag{3}$$

where ρ is resistivity, σ is conductivity, μ_n, and μ_p is the mobility of electron and hole, respectively, and n, p is the carrier concentration of

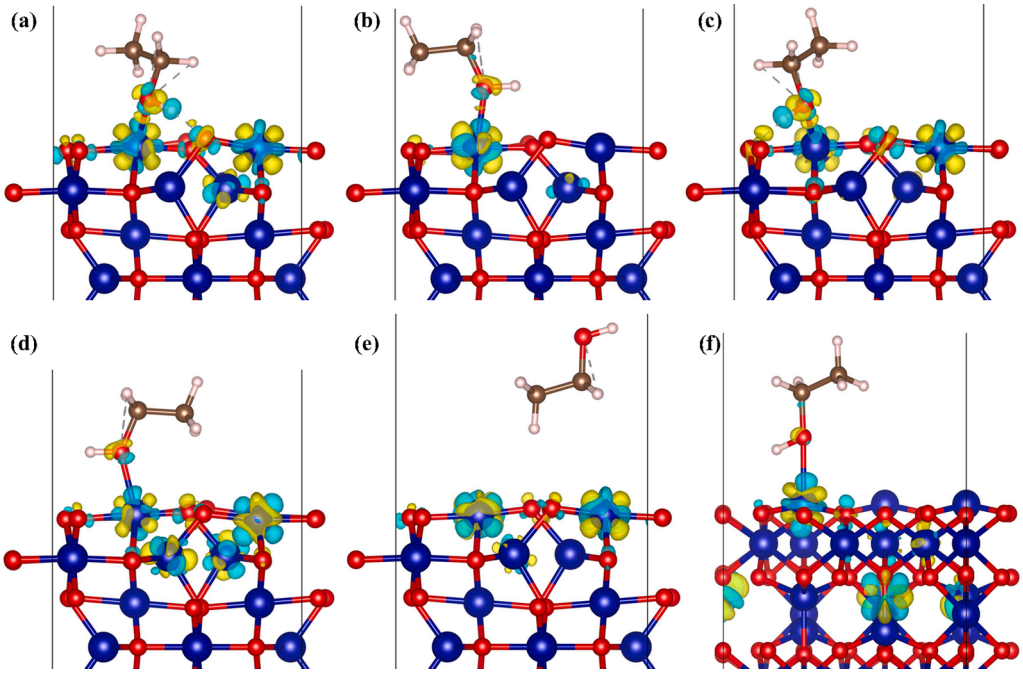

Fig. 10. (a-e) Calculated charges densities differences for ethanol adsorbed on the surface site S1-S5 of Co_3O_4(1 1 0). (f) Charges density differences for ethanol adsorbed on the surface of Co_3O_4(1 1 1). The blue and yellow areas represent depletion and accumulation of charges, respectively.

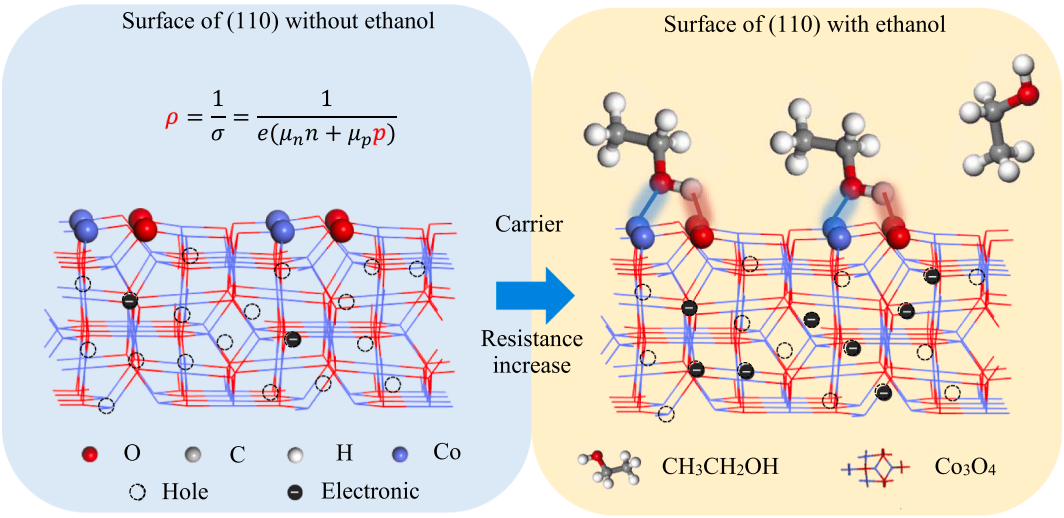

Fig. 11. Sensing mechanism of the Co_3O_4 nanorods and nanosheet sensors with ethanol gas.

electron and hole, respectively. When the Co_3O_4 sensor was exposed to ethanol gas, the exchange of electrons between ethanol molecules and the surface of the Co_3O_4 sensor leads to the recombination of carrier holes and electrons in the Co_3O_4 sensor. The recombination of electrons and holes in the Co_3O_4 materials will lead to a decrease in the carrier concentration, which increase the resistivity of ρ.

The high response value of the nanorod array of Co_3O_4 sensor to ethanol gas can be attributed to the exposed surface of Co_3O_4(1 1 0). The DFT calculations in this work revealed that the charge transfer of ethanol molecules adsorbed on the surface of Co_3O_4(1 1 0) exposed by the nanorods array was substantially larger than that of the surface of Co_3O_4(1 1 1) exposed by the nanosheets array. Moreover, the synergistic adsorption of ethanol molecules with the surface of Co_3O_4(1 1 0) exhibited the lowest adsorption energy, which indicated that the ethanol molecules were more likely to interact with the surface of Co_3O_4(1 1 0) exposed by the nanorods array. Opposite, the lower adsorption energy of ethanol molecules on the surface of Co_3O_4(1 1 1) exposed by the nanosheets array makes only a small amount of ethanol molecules exchange electrons with the sensor. Due to the synergistic adsorption effect of the ethanol gas with the nanorods arrays exposed on the surface of Co_3O_4(1 1 0), a large number of electrons recombine with the holes in the sensor of Co_3O_4 nanorod arrays, which leads to a decrease in the carrier concentration, so the change in resistivity of the nanorod arrays was much better than that of the nanosheets arrays exposed on the surface of Co_3O_4(1 1 1). This results in a high gas response of the nanorod arrays. Besides, owing to the synergistic adsorption effect between the alcohol hydroxyl group on the ethanol molecular and the metal cation and low-coordination oxygen atoms on the sensor surface, ethanol exhibits a higher response compared with other VOC gases. Considering methanol and isopropanol, their response values were very close. Due to the difference in the carbon chain structure of methanol, isopropanol and ethanol, their optimal working temperature may be different and thus the gas-sensing response of ethanol was higher than that of the others.

4. Conclusion

In summary, Co_3O_4 nanosheet (Co_3O_4-NS) with exposed (1 1 1) facets, and Co_3O_4 nanorods (Co_3O_4-NR) with exposed (1 1 0) facets in-situ grown on alumina ceramic tubes were prepared and studied in sensing of VOC gas. Experimental results suggested that compared with the Co_3O_4-NS, the ethanol gas-sensing performance and selectivity of Co_3O_4-NR were higher. Moreover, the adsorption configuration and electronic characteristics of ethanol molecules on the surface of Co_3O_4(1 1 0) were researched in detail by DFT. The calculation results of different adsorption models demonstrated that the oxygen and hydrogen atoms on the hydroxyl groups in ethanol have the lowest adsorption energies ($E_{ads} = -3.44$ eV) when they synergistic adsorbed with the Co atom and the indirectly adjacent low-coordinated oxygen atoms on the (1 1 0) surface. Due to the presence of the synergistic adsorption effects, the adsorption capability of the surface for hydroxyl groups was better than methyl groups in ethanol. Compared with the surface of Co_3O_4(1 1 1), Co_3O_4(1 1 0) has a large number of synergistic adsorption sites. Preparation of oxide semiconductor materials sensors with similar facets may contribute to improved sensing performance.

CRediT authorship contribution statement

Songlin Li: Project administration, Writing – review & editing. **Chen Wang:** Data curation, Investigation. **Zhaohui Lei:** Investigation, Methodology. **Shanfu Sun:** Formal analysis. **Jie Gao:** Software, Formal analysis. **Pengfei Cheng:** Investigation, Methodology. **Hai Wang:** Investigation, Formal analysis, Methodology.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgment

This work was supported by the National Natural Science Foundation of China (No. 22172119), Natural Science Basic Research Plan in Shaanxi Province of China (2021JM-047), National Key Research and Development Program (Nos. 2021YFC3090400 and 2021YFC3090402), Inner Mongolia Autonomous Region Science and Technology Plan Project (No. 2020GG0185, 2021GG0160 and 2022YFHH0073), High-performance Computing Platform of XiDian University.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.apsusc.2022.155776.

References

[1] T.M. Ngoc, N.V. Duy, N.D. Hoa, C.M. Hung, H. Nguyen, N.V. Hieu, Effective design and fabrication of low-power-consumption self-heated SnO_2 nanowire sensors for reducing gases, Sensor Actuat. B-Chem. 295 (2019) 144–152, https://doi.org/10.1016/j.snb.2019.05.074.

[2] Z. Meng, A. Aykanat, K.A. Mirica, Welding Metallophthalocyanines into Bimetallic Molecular Meshes for Ultrasensitive, Low-Power Chemiresistive Detection of Gases, J. Am. Chem. Soc. 141 (2019) 2046–2053, https://doi.org/10.1021/jacs.8b11257.

[3] Z. Meng, R.M. Stolz, K.A. Mirica, Two-Dimensional Chemiresistive Covalent Organic Framework with High Intrinsic Conductivity, J. Am. Chem. Soc. 141 (2019) 11929–11937, https://doi.org/10.1021/jacs.9b03441.

[4] M. Ko, L. Mendecki, A.M. Eagleton, C.G. Durbin, R.M. Stolz, Z. Meng, K.A. Mirica, Employing Conductive Metal-Organic Frameworks for Voltammetric Detection of Neurochemicals, J. Am. Chem. Soc. 142 (2020) 11717–11733, https://doi.org/10.1021/jacs.9b13402.

[5] A. Aykanat, Z. Meng, R.M. Stolz, C.T. Morrell, K.A. Mirica, Bimetallic Two-Dimensional Metal-Organic Frameworks for the Chemiresistive Detection of Carbon Monoxide, Angew. Chem.-Int. Edit. 61 (2022) e202113665.

[6] M. Jang, J. Lee, S.Y. Park, J. Lee, K.M. Lee, W. Song, S. Myung, S.S. Lee, H.-K. Jung, Y.C. Kang, S.K. Kwak, K.-S. An, Rational surface modification of ZnO with siloxane polymers for room-temperature-operated thin-film transistor-based gas sensors, Appl. Surf. Sci. 542 (2021), 148704, https://doi.org/10.1016/j.apsusc.2020.148704.

[7] Y. Wu, J.J. Gooding, The application of single molecule nanopore sensing for quantitative analysis, Chem. Soc. Rev. 51 (2022) 3862–3885, https://doi.org/10.1039/d1cs00988e.

[8] L. Fan, N. Xu, H. Chen, J. Zhou, S. Deng, A millisecond response and microwatt power-consumption gas sensor: Realization based on cross-stacked individual Pt-coated WO_3 nanorods, Sensor Actuat. B-Chem. 346 (2021), 130545, https://doi.org/10.1016/j.snb.2021.130545.

[9] H. Koga, K. Nagashima, Y. Huang, G. Zhang, C. Wang, T. Takahashi, A. Inoue, H. Yan, M. Kanai, Y. He, K. Uetani, M. Nogi, T. Yanagida, Paper-Based Disposable Molecular Sensor Constructed from Oxide Nanowires, Cellulose Nanofibers, and Pencil-Drawn Electrodes, Acs Appl. Mater. Inter. 11 (2019) 15044–15050, https://doi.org/10.1021/acsami.9b01287.

[10] X. Song, L. Li, X. Chen, Q. Xu, B. Song, Z. Pan, Y. Liu, F. Juan, F. Xu, B. Cao, Enhanced triethylamine sensing performance of alpha-Fe_2O_3 nanoparticle/ZnO nanorod heterostructures, Sensor Actuat. B-Chem. 298 (2019), 126917, https://doi.org/10.1016/j.snb.2019.126917.

[11] H. Du, W. Pu, C. Yang, Morphology control of Co_3O_4 with nickel incorporation for highly efficient oxygen evolution reaction, Appl. Surf. Sci. 541 (2021), 148221, https://doi.org/10.1016/j.apsusc.2020.148221.

[12] C.P. Plaisance, R.A. van Santen, Structure Sensitivity of the Oxygen Evolution Reaction Catalyzed by Cobalt(II, III) Oxide, J. Am. Chem. Soc. 137 (2015) 14660–14672, https://doi.org/10.1021/jacs.5b07779.

[13] T. Lee, M. Ferri, S. Piccinin, A. Selloni, Structure, Electronic Properties, and Defect Chemistry of Delafossite $CuRhO_2$ Bulk and Surfaces, Chem. Mater. 34 (2022) 1567–1577, https://doi.org/10.1021/acs.chemmater.1c03248.

[14] Y. Xu, F. Zhang, T. Sheng, T. Ye, D. Yi, Y. Yang, S. Liu, X. Wang, J. Yao, Clarifying the controversial catalytic active sites of Co_3O_4 for the oxygen evolution reaction, J. Mater. Chem. A. 7 (2019) 23191–23198, https://doi.org/10.1039/C9TA08379K.

[15] P. Cheng, F. Dang, Y. Wang, J. Gao, L. Xu, C. Wang, L. Lv, X. Li, B. Zhang, B. Liu, Gas sensor towards n-butanol at low temperature detection: Hierarchical flower-like Ni-doped Co_3O_4 based on solvent-dependent synthesis, Sensor Actuat. B-Chem. 328 (2021), 129028, https://doi.org/10.1016/j.snb.2020.129028.

[16] Y. Xing, L.-X. Zhang, H. Xu, Y.-Y. Yin, M.-X. Chong, L.-J. Bie, Defect-rich ultrathin Sn_2O_3 nanosheets with dominant polar (100) facets for efficient gas and humidity sensor applications, Sensor Actuat. B-Chem. 349 (2021), 130816, https://doi.org/10.1016/j.snb.2021.130816.

[17] M. Liu, L. Piao, L. Zhao, S. Ju, Z. Yan, T. He, C. Zhou, W. Wang, Anatase TiO2 single crystals with exposed 001 and 110 facets: facile synthesis and enhanced photocatalysis, Chem. Commun. 46 (2010) 1664–1666, https://doi.org/10.1039/B924172H.

[18] Y. Zhao, F. Pan, H. Li, T. Niu, G. Xu, W. Chen, Facile synthesis of uniform α-Fe_2O_3 crystals and their facet-dependent catalytic performance in the photo-Fenton reaction, J. Mater. Chem. A 1 (2013) 7242–7246, https://doi.org/10.1039/C3TA10966F.

[19] W. Gu, F. Teng, Z. Yang, Z. Liu, Surface electrochemistry of truncated Cu_2O octahedrons, Mater. Chem. Phys. 263 (2021), 124415, https://doi.org/10.1016/j.matchemphys.2021.124415.

[20] X. Gao, T. Zhang, An overview: Facet-dependent metal oxide semiconductor gas sensors, Sensor Actuat. B-Chem. 277 (2018) 604–633, https://doi.org/10.1016/j.snb.2018.08.129.

[21] V.R. Goncales, J. Lian, S. Gautam, D. Hagness, Y. Yang, R.D. Tilley, S. Ciampi, J. J. Gooding, Heterojunctions Based on Amorphous Silicon: A Versatile Surface Engineering Strategy To Tune Peak Position of Redox Monolayers on Photoelectrodes, J. Phys. Chem. C 124 (2020) 836–844, https://doi.org/10.1021/acs.jpcc.9b11252.

[22] A.R. Poerwoprajitno, L. Gloag, J. Watt, S. Cheong, X. Tan, H. Lei, H.A. Tahini, A. Henson, B. Subhash, N.M. Bedford, B.K. Miller, P.B. O'Mara, T.M. Benedetti, D. L. Huber, W. Zhang, S.C. Smith, J.J. Gooding, W. Schuhmann, R.D. Tilley, A single-Pt-atom-on-Ru-nanoparticle electrocatalyst for CO-resilient methanol oxidation, Nat. Catal. 5 (2022) 231–237, https://doi.org/10.1038/s41929-022-00756-9.

[23] N. Zhang, Y. Fan, Y. Lu, C. Li, J. Zhou, X. Li, S. Adimi, S. Ruan, Synthesis of au-decorated SnO_2 crystallites with exposed (221) facets and their enhanced acetylene sensing properties, Sensor Actuat. B-Chem. 307 (2020), 127629, https://doi.org/10.1016/j.snb.2019.127629.

[24] Y. Ren, J. Yang, Y. Ma, B. Liu, X. Zhang, L. Wang, Y. Yuan, J. Liu, M. Wang, Q. Du, H. Zhao, H. Yang, Increasing sensing sensitivity of the Fe-alpha-Fe_2O_3 (104) surface by hydrogenation and the sensing reaction molecule mechanism, Sensor Actuat. B-Chem. 281 (2019) 366–374, https://doi.org/10.1016/j.snb.2018.10.088.

[25] L. Zhang, J. Zhang, Y. Huang, H. Xu, X. Zhang, H. Lu, K. Xu, P.K. Chu, F. Ma, Stability and Sensing Enhancement by Nanocubic CeO2 with 100 Polar Facets on Graphene for NO2 at Room Temperature, Acs Appl. Mater. Inter. 12 (2020) 4722–4731, https://doi.org/10.1021/acsami.9b18155.

[26] H. Bi, L.-X. Zhang, Y. Xing, P. Zhang, J.-J. Chen, J. Yin, L.-J. Bie, Morphology-controlled synthesis of CeO_2 nanocrystals and their facet-dependent gas sensing properties, Sensor Actuat. B-Chem. 330 (2021), 129374, https://doi.org/10.1016/j.snb.2020.129374.

[27] L. Zhang, J. Shi, Y. Huang, H. Xu, K. Xu, P.K. Chu, F. Ma, Octahedral SnO_2/Graphene Composites with Enhanced Gas-Sensing Performance at Room Temperature, Acs Appl. Mater. Inter. 11 (2019) 12958–12967, https://doi.org/10.1021/acsami.8b22533.

[28] H.-S. Chen, T.M. Benedetti, J. Lian, S. Cheong, P.B. O'Mara, K.O. Sulaiman, C.H. W. Kelly, R.W.J. Scott, J.J. Gooding, R.D. Tilley, Role of the Secondary Metal in Ordered and Disordered Pt-M Intermetallic Nanoparticles: An Example of Pt3Sn Nanocubes for the Electrocatalytic Methanol Oxidation, ACS Catal. 11 (2021) 2235–2243, https://doi.org/10.1021/acscatal.0c05370.

[29] R. Wu, F. Tian, Z. Liu, X. Xue, J. Zhang, J. Zu, CH_4 activation and sensing on hexagonal WO_3 (001) and (110) surfaces, Appl. Surf. Sci. 481 (2019) 1154–1159, https://doi.org/10.1016/j.apsusc.2019.03.094.

[30] B. Liu, L. Wang, Y. Ma, Y. Yuan, J. Yang, M. Wang, J. Liu, X. Zhang, Y. Ren, Q. Du, H. Zhao, C. Pei, S. Liu, H. Yang, Enhanced gas-sensing properties and sensing mechanism of the foam structures assembled from NiO nanoflakes with exposed 111 facets, Appl. Surf. Sci. 470 (2019) 596–606, https://doi.org/10.1016/j.apsusc.2018.11.129.

[31] H. Liu, F. Wang, K. Hu, T. Li, Y. Yan, Pd_4 cluster decorated SnO_2 nanowire for detecting characteristic gases in oil-immersed transformers: A theoretical and experimental study, Appl. Surf. Sci. 590 (2022), 153122, https://doi.org/10.1016/j.apsusc.2022.153122.

[32] H. Cui, J. Jiang, C. Gao, F. Dai, J. An, Z. Wen, Y. Liu, DFT study of Cu-modified and Cu-embedded WSe_2 monolayers for cohesive adsorption of NO_2, SO_2, CO_2, and H_2S, Appl. Surf. Sci. 583 (2022), 152522, https://doi.org/10.1016/j.apsusc.2022.152522.

[33] S. Gao, Z. Ma, C. Xiao, W. Du, X. Sun, Q. Li, R. Sa, C. Sun, High-Throughput computational screening of Single-atom embedded in defective BN nanotube for electrocatalytic nitrogen fixation, Appl. Surf. Sci. 591 (2022), 153130, https://doi.org/10.1016/j.apsusc.2022.153130.

[34] B. Lalmuanchhana, R.C.T. Lalroliana, R.M. Lalhriatzuala, Transition metal decorated ZnO monolayer for CO and NO sensing: A DFT + U study with vdW correction, Appl. Surf. Sci. 604 (2022), 154570, https://doi.org/10.1016/j.apsusc.2022.154570.

[35] S.U.D. Shamim, D. Roy, S. Alam, A.A. Piya, M.S. Rahman, M.K. Hossain, F. Ahmed, Doubly doped graphene as gas sensing materials for oxygen-containing gas molecules: A first-principles investigation, Appl. Surf. Sci. 596 (2022), 153603, https://doi.org/10.1016/j.apsusc.2022.153603.

[36] W.T. Yang, B. Zhao, C.Y. Li, P. Guo, M. Li, X.H. Ge, M. Zhang, X.Y. Guan, J.J. Wang, Enhanced gas molecules adsorption on γ-graphyne doped with Fe atom: A first-principles study, Appl. Surf. Sci. 601 (2022), 154083, https://doi.org/10.1016/j.apsusc.2022.154083.

[37] Y. Yong, R. Gao, X. Yuan, Z. Zhao, S. Hu, Y. Kuang, Gas sensing and capturing based on the C_7N_6 monolayer with and without metal decoration: A first-principles investigation, Appl. Surf. Sci. 591 (2022), 153129, https://doi.org/10.1016/j.apsusc.2022.153129.

[38] A. Gurlo, Nanosensors: towards morphological control of gas sensing activity. SnO_2, In_2O_3, ZnO and WO_3 case studies, Nanoscale 3 (2011) 154–165, https://doi.org/10.1039/c0nr00560f.

[39] J. Xu, Z. Xue, N. Qin, Z. Cheng, Q. Xiang, The crystal facet-dependent gas sensing properties of ZnO nanosheets: Experimental and computational study, Sensor Actuat. B-Chem. 242 (2017) 148–157, https://doi.org/10.1016/j.snb.2016.09.193.

[40] Q. Li, D. Chen, J. Miao, S. Lin, Z. Yu, D. Cui, Z. Yang, X. Chen, Highly sensitive sensor based on ordered porous ZnO nanosheets for ethanol detecting application, Sensor Actuat. B-Chem. 326 (2021), 128952, https://doi.org/10.1016/j.snb.2020.128952.

[41] J. Ma, Y. Ren, X. Zhou, L. Liu, Y. Zhu, X. Cheng, P. Xu, X. Li, Y. Deng, D. Zhao, Pt Nanoparticles Sensitized Ordered Mesoporous WO_3 Semiconductor: Gas Sensing Performance and Mechanism Study, Adv. Funct. Mater. 28 (2018) 1705268, https://doi.org/10.1002/adfm.201705268.

[42] M.R. Alenezi, S.J. Henley, N.G. Emerson, S.R.P. Silva, From 1D and 2D ZnO nanostructures to 3D hierarchical structures with enhanced gas sensing properties, Nanoscale 6 (2014) 235–247, https://doi.org/10.1039/c3nr04519f.

[43] N. Joshi, T. Hayasaka, Y. Liu, H. Liu, O.N. Oliveira, L. Lin, A review on chemiresistive room temperature gas sensors based on metal oxide nanostructures, graphene and 2D transition metal dichalcogenides, Microchim. Acta 185 (2018) 213, https://doi.org/10.1007/s00604-018-2750-5.

[44] U.T. Nakate, P. Bhuyan, Y.T. Yu, S. Park, Synthesis and characterizations of highly responsive H_2S sensor using p-type Co_3O_4 nanoparticles/nanorods mixed

nanostructures, Int. J. Hydrogen Energ. 47 (2022) 8145–8154, https://doi.org/10.1016/j.ijhydene.2021.12.115.
[45] H.-J. Kim, J.-H. Lee, Highly sensitive and selective gas sensors using p-type oxide semiconductors: Overview, Sensor Actuat. B-Chem. 192 (2014) 607–627, https://doi.org/10.1016/j.snb.2013.11.005.
[46] R. Zhang, S. Gao, T. Zhou, J. Tu, T. Zhang, Facile preparation of hierarchical structure based on p-type Co_3O_4 as toluene detecting sensor, Appl. Surf. Sci. 503 (2020), 144167, https://doi.org/10.1016/j.apsusc.2019.144167.
[47] X. Fan, Y. Xu, C. Ma, W. He, In-situ growth of Co_3O_4 nanoparticles based on electrospray for an acetone gas sensor, J. Alloy Compd. 854 (2021), 157234, https://doi.org/10.1016/j.jallcom.2020.157234.
[48] Y. Jian, M. Tian, C. He, J. Xiong, Z. Jiang, H. Jin, L. Zheng, R. Albilali, J.-W. Shi, Efficient propane low-temperature destruction by Co_3O_4 crystal facets engineering: Unveiling the decisive role of lattice and oxygen defects and surface acid-base pairs, Appl. Catal. B-Environ. 283 (2021), 119657, https://doi.org/10.1016/j.apcatb.2020.119657.
[49] H. Sun, H. Lv, Facet engineering of Nano-Co_3O_4 for catalytic and gas sensor performance: A mechanism insight, J. Alloy Compd. 823 (2020), 153742, https://doi.org/10.1016/j.jallcom.2020.153742.
[50] A. Aykanat, C.G. Jones, E. Cline, R.M. Stolz, Z. Meng, H.M. Nelson, K.A. Mirica, Conductive Stimuli-Responsive Coordination Network Linked with Bismuth for Chemiresistive Gas Sensing, ACS Appl. Mater. Interfaces 13 (2021) 60306–60318, https://doi.org/10.1021/acsami.1c14453.
[51] Q. Yang, Z. Lu, Z. Chang, W. Zhu, J. Sun, J. Liu, X. Sun, X. Duan, Hierarchical Co_3O_4 nanosheet@nanowire arrays with enhanced pseudocapacitive performance, RSC Adv. 2 (2012) 1663–1668, https://doi.org/10.1039/C1RA01008E.
[52] C. Wang, Y. Wang, P. Cheng, L. Xu, F. Dang, T. Wang, Z. Lei, In-situ generated TiO_2/α-Fe_2O_3 heterojunction arrays for batch manufacturing of conductometric acetone gas sensors, Sensor Actuat. B-Chem. 340 (2021), 129926, https://doi.org/10.1016/j.snb.2021.129926.
[53] G. Kresse, J. Furthmüller, Efficient iterative schemes for ab initio total-energy calculations using a plane-wave basis set, Phys. Rev. B 54 (1996) 11169–11186, https://doi.org/10.1103/PhysRevB.54.11169.
[54] G. Kresse, J. Furthmüller, Efficiency of ab-initio total energy calculations for metals and semiconductors using a plane-wave basis set, Comp. Mater. Sci. 6 (1996) 15–50, https://doi.org/10.1016/0927-0256(96)00008-0.
[55] J. Ren, K. Song, Z. Li, Q. Wang, J. Li, Y. Wang, D. Li, C.K. Kim, Activation of formyl C-H and hydroxyl O-H bonds in HMF by the CuO (111) and Co_3O_4 (110) surfaces: A DFT study, Appl. Surf. Sci. 456 (2018) 174–183, https://doi.org/10.1016/j.apsusc.2018.06.120.
[56] J.P. Perdew, K. Burke, M. Ernzerhof, Generalized Gradient Approximation Made Simple, Phys. Rev. Lett. 77 (1996) 3865–3868, https://doi.org/10.1103/PhysRevLett.77.3865.
[57] V.N. Staroverov, G.E. Scuseria, J.P. Perdew, E.R. Davidson, J. Katriel, High-density limit of the Perdew-Burke-Ernzerhof generalized gradient approximation and related density functionals, Phys. Rev. A 74 (2006), 044501, https://doi.org/10.1103/PhysRevA.74.044501.
[58] V. Singh, M. Kosa, K. Majhi, D.T. Major, Putting DFT to the Test: A First-Principles Study of Electronic, Magnetic, and Optical Properties of Co_3O_4, J. Chem. Theory Comput. 11 (2015) 64–72, https://doi.org/10.1021/ct500770m.
[59] J. Liu, V. Fung, Y. Wang, K. Du, S. Zhang, L. Nguyen, Y. Tang, J. Fan, D. Jiang, F. F. Tao, Promotion of catalytic selectivity on transition metal oxide through restructuring surface lattice, Appl. Catal. B-Environ. 237 (2018) 957–969, https://doi.org/10.1016/j.apcatb.2018.05.013.
[60] V. Wang, N. Xu, J.-C. Liu, G. Tang, W.-T. Geng, VASPKIT: A user-friendly interface facilitating high-throughput computing and analysis using VASP code, Comput. Phys. Commun. 267 (2021), 108033, https://doi.org/10.1016/j.cpc.2021.108033.
[61] K. Shojaee, B.S. Haynes, A. Montoya, The catalytic oxidation of NH_3 on Co_3O_4(110): A theoretical study, P Combust. Inst. 36 (2017) 4365–4373, https://doi.org/10.1016/j.proci.2016.06.100.
[62] S. Wang, C. Zhao, S. Li, Y. Sun, First principles prediction of CH4 reactivities with Co_3O_4 nanocatalysts of different morphologies, Phys. Chem. Chem. Phys. 19 (2017) 30874–30882, https://doi.org/10.1039/c7cp04516f.
[63] W. Li, Y. Wang, X.Y. Cui, S. Yu, Y. Li, Y. Hu, M. Zhu, R. Zheng, S.P. Ringer, Crystal Facet Effects on Nanomagnetism of Co_3O_4, Acs Appl. Mater. Inter. 10 (2018) 19235–19247, https://doi.org/10.1021/acsami.8b03934.
[64] S.V. Gallego, J.M. Perez-Mato, L. Elcoro, E.S. Tasci, R.M. Hanson, M.I. Aroyo, G. Madariaga, MAGNDATA: towards a database of magnetic structures. II. The incommensurate case, J. Appl. Crystallogr. 49 (2016) 1941–1956, https://doi.org/10.1107/S1600576716015491.
[65] W.L. Roth, The magnetic structure of Co_3O_4, J. Phys. Chem. Solids 25 (1964) 1–10, https://doi.org/10.1016/0022-3697(64)90156-8.
[66] M. Yu, D.R. Trinkle, Accurate and efficient algorithm for Bader charge integration, J. Chem. Phys. 134 (6) (2011), https://doi.org/10.1063/1.3553716.